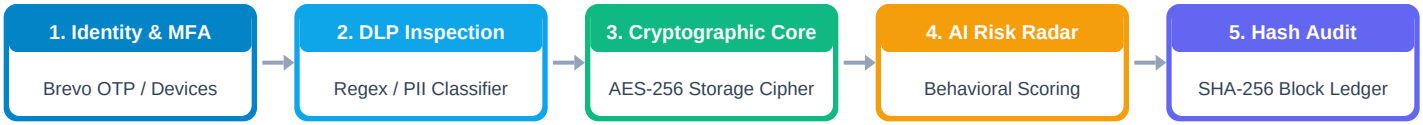
Smart Cloud Shield (SCS)

Enterprise Cloud Security Architecture & Comprehensive Operations Guide

1. Executive Summary: What is Smart Cloud Shield?

Smart Cloud Shield (SCS) is an intelligent, zero-trust cloud file storage and access governance platform. Standard cloud drives (like basic Dropbox or S3 buckets) implicitly trust any authenticated session; if user credentials are leaked or an insider gains direct storage access, files are exposed. SCS enforces a **Defense-in-Depth model** where every access request is validated by dynamic Multi-Factor Authentication, scanned for sensitive data patterns (DLP), encrypted at rest before hitting physical disks, and continuously evaluated by a real-time behavioral AI risk radar.

Figure 1: The 5-Stage Zero-Trust Security Pipeline



Pillar	Core Technology	Operational Capability	Target Threat Mitigated
Authentication	Brevo HTTPS API (Port 443)	Dynamic 6-digit OTP (10 min TTL)	Credential stuffing, password spraying
Data at Rest	Fernet / AES-CBC Symmetric Cipher	Envelope encryption + per-file wrapped DEK	Physical server compromise, storage theft
DLP Scanner	Verhoeff checksum + Regex Classifier	Detects Aadhaar, PAN, Credit Cards, RSA keys	Accidental data leaks, PII compliance breach
AI Risk Radar	Isolation Forest ML + Gemini LLM	Real-time 0–100 risk scoring & incident flagging	Insider exfiltration, off-hours brute force
Audit Ledger	SHA-256 Hash-Chaining	Block-style immutable log verification	Log tampering, non-repudiation evasion

2. Deep-Dive: The 8 Modular Components of SCS

Module 1: Landing Page & Live Pipeline Visualizer

The entry portal features an interactive 5-stage security pipeline and a real-time SVG risk telemetry gauge. Users can review platform capabilities and jump directly into the demo or authentication portal.

Module 2: Authentication & Device-Fingerprinted MFA

Implements salted password authentication coupled with dynamic 6-digit One-Time Passwords delivered via the **Brevo REST API** over HTTPS port 443 (bypassing cloud provider SMTP port 587 firewall blocks). It inspects browser telemetry to identify known vs. untrusted devices, increasing the risk score for unknown hardware.

Module 3: Secure Cloud Drive & Sensitive Data Inspection (DLP)

Users upload and manage files with instant Data Loss Prevention (DLP) classification. The scanner evaluates Aadhaar numbers using the official Verhoeff checksum algorithm, PAN patterns, credit card formats, and private SSH/API keys. Files are classified as Public, Internal, Confidential, or Highly Confidential with a 0–100 sensitivity score.

Module 4: Storage Encryption at Rest & Secure Deduplication

Raw file payloads are never written as plaintext. The platform generates an envelope key, encrypts the byte stream using symmetric ciphers, and stores ciphertext on disk. When identical files are uploaded, cryptographic SHA-256 deduplication saves disk space while maintaining independent ownership and access control records.

Module 5: SOC Security Operations & Live Telemetry Dashboard

The centralized administrative dashboard monitors 24-hour telemetry: Allowed Downloads, Warned Accesses, Blocked Exfiltrations, and Open Incidents. Includes interactive outcome charts, risk trend graphs, and an instant test email dispatcher.

Module 6: Autonomous Incident Management & Threat Quarantine

When high-risk actions occur (e.g. guest accessing privileged credentials from an unknown IP), the system raises a security incident. Security analysts can review severity scores, examine causal reasons, inspect quarantine states, and mark incidents as Investigating, Resolved, or False Positive.

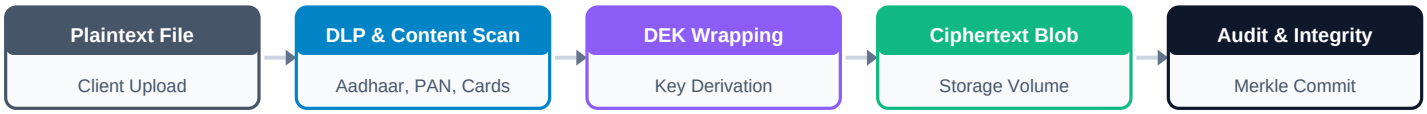
Module 7: Role-Based Access Control (RBAC) & Governance

Enforces two distinct tiers: Standard Employees (Alice, Bob, Carol) restricted to their own department files, and Administrators with elevated privileges for system-wide auditing, user account modifications, and threat resolution.

Module 8: Cryptographic Audit Ledger & File Integrity Verification

Every security event is appended to an immutable, hash-chained audit log where each block's cryptographic hash commits to the previous entry. An automated integrity verification routine recalculates SHA-256 hashes across stored blobs to catch unauthorized tampering.

Figure 2: Cryptographic Storage & Encryption at Rest Lifecycle



3. Step-by-Step User Guide & Live Demo Walkthrough

Follow these workflows to demonstrate and operate the live application on Render or locally. Use the credentials table below to test different roles and clearance levels.

Username	Password	Role & Clearance	Registered Email (MFA Destination)	Demo Scenario
admin	Pass@12345	Admin (Level 3)	banagarsankeerth@gmail.com	Full SOC Dashboard, Incidents, Audit Logs
alice	Pass@12345	Employee - Finance (L3)	raghavendranr28781@gmail.com	Owns payroll and customer card data
bob	Pass@12345	Employee - HR (L2)	raghavendranr1218@gmail.com	Owns HR contacts and meeting notes
carol	Pass@12345	Employee - Eng (L3)	carol@corp.example	Owns server credentials & infrastructure files
guest1	Pass@12345	Guest - General (L0)	guest1@corp.example	Untrusted actor used to trigger exfiltration blocks

Workflow A: User Authentication & Live MFA Verification

- 1. Navigate to the login portal (/login) and enter credentials (e.g. alice / Pass@12345).
- 2. The server verifies password hash, generates a secure 6-digit OTP, and dispatches it via Brevo's HTTPS API directly to the user's registered inbox.
- 3. The browser redirects to the OTP verification screen (/otp).
- 4. Check the inbox (or terminal in local mode), enter the 6-digit code, and click **Verify Code** to complete login.

Workflow B: Encrypted File Upload & Automated DLP Tagging

- 1. From the top navigation bar, click **Files** to access the storage repository.
- 2. Click **Choose File**, select a sample document (e.g. CSV or TXT with sensitive data), and click **Upload & Encrypt**.
- 3. SCS executes DLP regex and Verhoeff checks, categorizes the sensitivity, encrypts the payload with Fernet symmetric cipher, and stores the ciphertext on disk.
- 4. To retrieve the file, click **Download**. The system validates the clearance level, decrypts the bytes in-memory, and streams the decrypted file.

Workflow C: SOC Telemetry Monitoring & Test Dispatch Delivery

- 1. Log in as admin and click **Dashboard** in the top navigation bar.
- 2. Review the top telemetry cards: Allowed Downloads, Warned Requests, Blocked Exfiltrations, and Open Incidents.
- 3. Review the **Download Outcomes** bar chart and the **Risk Trend** progression graph.
- 4. In the **Alert Dispatches** card, type your email address and click **Send Test** to verify real-time email delivery via Brevo.

Workflow D: Incident Management & Threat Mitigation

- 1. In the Admin navbar, click **Incidents** to inspect all flagged security events.
- 2. Click on any active incident to review the causal breakdown (e.g. unknown IP, off-hours access, privilege escalation).
- 3. Use the status control to transition the incident from **Open** to **Investigating** or **Resolved**.

Workflow E: Cryptographic Audit & File Integrity Verification

- 1. Click **Audit** in the navbar to inspect the hash-chained ledger. Each row displays the previous hash and the current SHA-256 digest.
- 2. Click **Integrity** in the navbar. Click **Run Verification Check** to scan all stored ciphertext blobs against their original hashes.
- 3. The system reports 100% integrity match or flags any simulated file tampering.

Workflow F: AI Threat Insights Hub

- 1. In the Admin navbar, click **AI Insights**.
- 2. Click **Generate 24h Summary**. The Google Gemini AI model analyzes recent telemetry, incident counts, and audit logs to output a synthesized executive threat briefing.

4. Essential Technical Terminologies & Exam Reference

• **Zero-Trust Architecture:** A cybersecurity paradigm based on 'never trust, always verify'. Every transaction, user, and device is continually authenticated regardless of network location.

• **Multi-Factor Authentication (MFA):** Security control requiring two or more independent credentials for access: something you know (password) and something you have (time-limited email OTP).

• **Encryption at Rest:** Cryptographic protection of stored files on physical storage disks so data cannot be extracted even if the underlying volume is stolen or inspected.

• **Envelope Encryption (DEK / KEK):** A two-tier cryptographic hierarchy where files are encrypted using a Data Encryption Key (DEK), which is in turn wrapped by a Master Key Encryption Key (KEK).

• **Data Loss Prevention (DLP):** Automated content inspection using patterns and mathematical checksums (e.g. Verhoeff algorithm) to identify and restrict sensitive data (Aadhaar, PAN, PCI cards).

• **Hash-Chained Audit Ledger:** A sequence of tamper-evident log records where each entry incorporates the cryptographic SHA-256 digest of the prior entry, creating an immutable blockchain-style audit history.

• **Role-Based Access Control (RBAC):** Restricting system access and capabilities according to authorized enterprise roles (Admin, Employee, Guest) and numerical clearance levels.

• **Telemetry Risk Index (0–100):** A normalized composite metric synthesized from contextual signals (IP novelty, device trust, failed login velocity, request time) to dynamically detect attacks.

Smart Cloud Shield Platform • End-to-End System Documentation & Operations Manual • Ready for Vivas, Demos, and Reviews